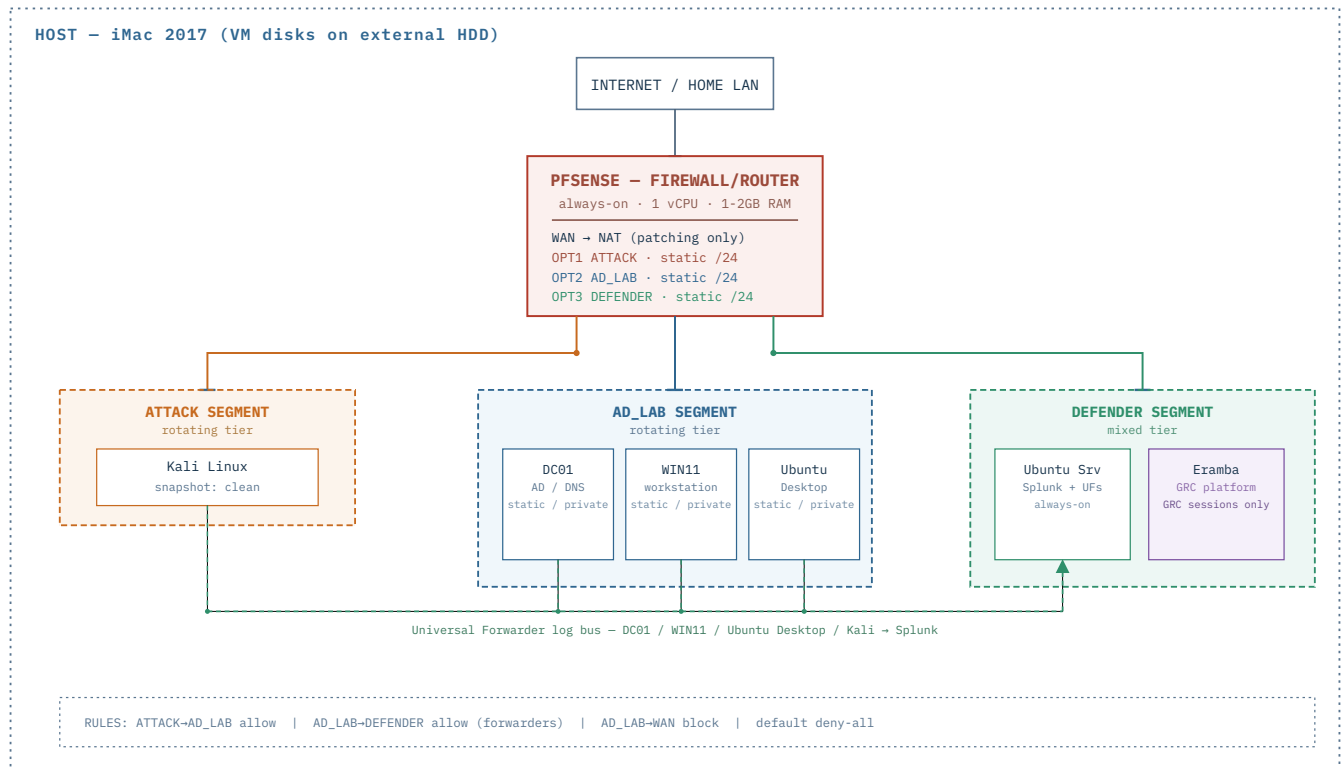


HOMELAB.LOCAL – NETWORK ARCHITECTURE

iMac 2017 · Intel i7 4C/8T · 48GB RAM · VirtualBox · Internal-
Network Segmentation

REV 2.0 –
PHASE 2



■ pfSense (always-on) ■ ATTACK (rotating) ■ AD_LAB (rotating)

■ DEFENDER – Splunk (always-on) ■ DEFENDER – Eramba (rotating, GRC only)

Note: VirtualBox Internal Networks stand in for VLAN-tagged bridges here – functionally equivalent isolation and routing-through-firewall behavior, without requiring a VLAN-aware hypervisor. Documented as a deliberate simplification, not an oversight.

DESIGN DECISIONS & RATIONALE

01 Is Kali connected to Splunk?

Yes, but this is a deliberate lab-only choice, not a production analog – real SOC's don't run defensive agents on the adversary's own machine. It's connected here specifically so I can cross-check the exact timestamp of an attack I launched against the exact timestamp Splunk detected it on the target

side, which validates detection timing accuracy. This forwarder is treated as optional in practice: it's useful for verification, not something a real enterprise would deploy.

02 Why does AD_LAB have no route to WAN?

Blocking AD_LAB → WAN closes the exfiltration/C2 pathway a compromised domain host would use in a real breach, and forces patching through a controlled, explicit process instead of ad hoc internet access. This is the segmentation rule doing actual work, not just existing on paper.

03 Why is Eramba rotating instead of always-on?

The host is a 4-core/8-thread machine — every always-on VM is permanent overhead. GRC work (risk register entries, control mapping, policy writing) doesn't need to run concurrently with attack generation, so Eramba only boots for its own sessions. This keeps the SOC skill-building sessions at the same resource footprint they were tuned for.

04 Why Internal Networks instead of true VLANs?

VirtualBox doesn't do VLAN-aware bridging cleanly the way Proxmox or a managed switch would. Internal Networks per segment give the same isolation and routing-through-pfSense behavior without needing a hypervisor this lab doesn't run. Named explicitly as a trade-off, since being asked "why not real VLANs" in an interview is a near-certainty.

05 Why do Splunk and Eramba share the DEFENDER segment instead of separate zones?

Both are oversight/evidence tools, not attack sources or targets — they belong at the same trust level. Splitting them would also cost a 5th pfSense interface, and VirtualBox's GUI natively supports 4 (WAN + 3 OPT) before requiring CLI workarounds. Grouping them by trust level rather than by function kept the design inside that constraint without losing anything meaningful.

06 Why default-deny instead of default-allow with specific blocks?

Default-deny is the least-privilege standard (and the CIS/NIST-expected posture) — every permitted path is an explicit, intentional rule rather than something that works until someone remembers to block it. It also means every allow rule in this lab was written and justified on purpose, which is real firewall-rule practice rather than an afterthought.

PROJECT

homelab.local — SOC/GRC Lab, Phase 2

DRAWN

homelab.local build log

DATE / REV

Jul 2026 — Rev 2.0